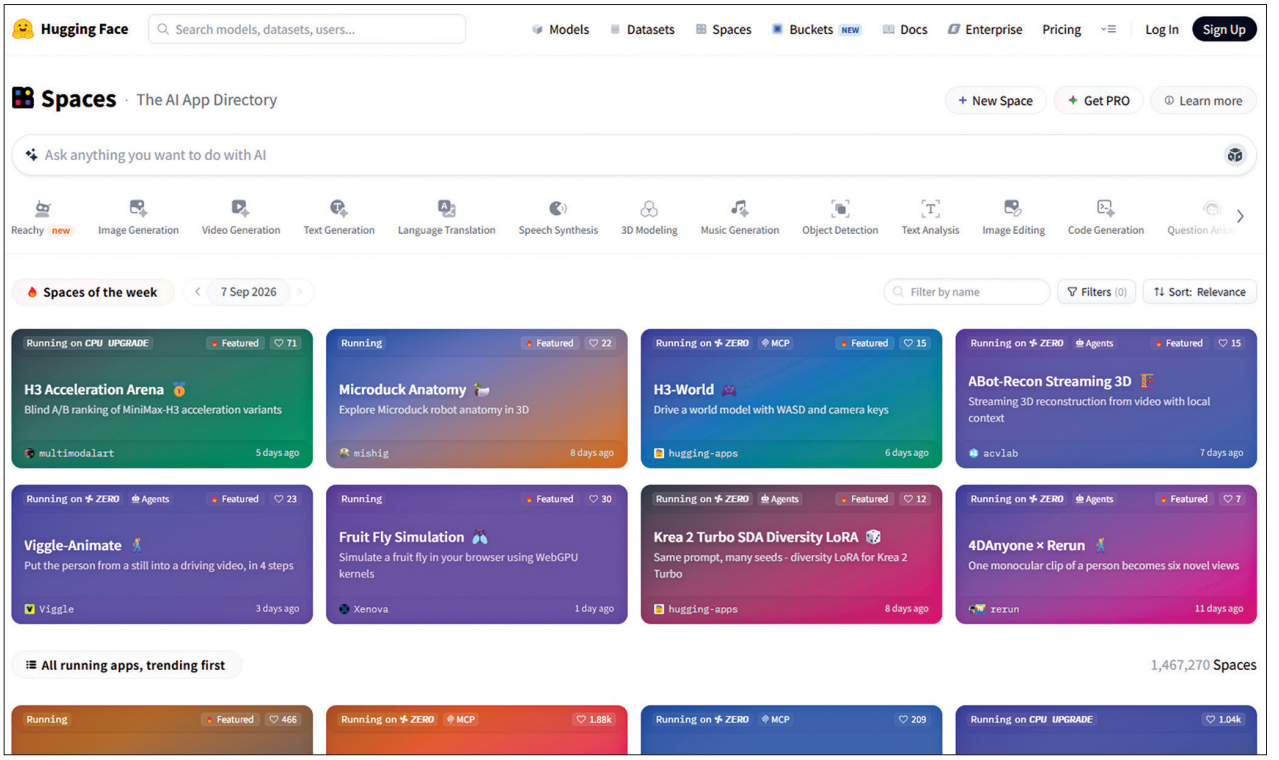


will encounter most often are Models, Datasets and Spaces. A Model repository contains an AI model and the information needed to use it: weights, configuration files, tokenizer files, a README, licence details and sometimes evaluation results.

A Dataset repository contains data. Datasets matter because model cards may tell you which ones were used for training or evaluation.

A Space is an interactive AI application or demo hosted through Hugging Face. It can be useful for trying a model before downloading it, although a Space may use different hardware or settings from your future local setup.

Hugging Face provides full-text search across models, datasets and Spaces and lets you restrict results to a category. For this chapter, stay primarily in Models.

Step 1: Search by task before hype

Suppose you want a local model for everyday writing. Typing only the name of the hottest model family into search is backwards. Start with the task.

Open Models and use the filters. Hugging Face metadata lets models be categorised by tasks such as text generation, image-to-text, automatic speech recognition and text-to-speech. Authors can also declare a `pipeline_tag`, which identifies the intended task and helps the Hub surface relevant interfaces.

For a local chatbot, you are usually looking for a text-generation or instruction model. For screenshots and photographs, look for image-text-to-text or the relevant multimodal category. For embeddings, search specifically for embedding or feature-extraction models rather than downloading a chat LLM.

Then narrow further. Filter by language if your needs are specific. Look for model families you shortlisted in Chapter 3. Pay attention to parameter count. Hugging Face's current tooling can even filter repositories by parameter range.

Think:

Task → size → language → runtime → model.

Not:

Popular model → download → discover it does not fit.

Step 2: Identify the original repository

This is one of the most important Hugging Face habits.

Imagine searching for a hypothetical `Acme-12B-Instruct`. You might find:

- `AcmeAI/Acme-12B-Instruct`
- `SomeUser/Acme-12B-Instruct-GGUF`
- `FastModels/Acme-12B-Instruct-Q4`
- `AnotherUser/Acme-12B-Instruct-Hindi`
- `AcmeFans/Acme-12B-Uncensored`

These are not interchangeable.

The original repository, usually published by the model developer or organisation, should be your first stop even if you ultimately download a community conversion. It is where you are most likely to find the authoritative model card, licence, intended use, architecture and context length.